Davronbek Malikov

LLM Engineer | RAG Engineer | Agentic AI Engineer | ML Engineer
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Summary

- LLM Engineer, RAG Engineer, Agentic AI Engineer, and ML Engineer with a Ph.D. in AI Convergence Engineering. I build LLM systems end to end: fine-tuning and inference serving, retrieval, agent orchestration, evaluation, and deployment on cloud and on-premise infrastructure.
- **LLM Engineering:** Supervised fine-tuning and parameter-efficient adaptation (LoRA/QLoRA), preference optimization (DPO), quantization (AWQ, GPTQ, GGUF), and local inference serving (vLLM, TGI, Ollama, llama.cpp).
- **RAG Engineering:** Standard RAG and GraphRAG pipelines with hybrid search (dense vectors + sparse BM25), cross-encoder reranking, hierarchical chunking, and vector databases (pgvector, Qdrant, Milvus).
- **Agentic AI Engineering:** Multi-agent architectures (Supervisor, Router, Network), plan-and-execute and self-correction loops, human-in-the-loop workflows, and tool use through MCP and function calling. Currently leading the AI engineering team at TripleH, shipping agents for legal intelligence, financial analysis, meeting automation, and enterprise digital twins.
- **ML Engineering:** Supervised and unsupervised machine learning on tabular data and Computer Vision, grounded in advanced mathematics, with 5 first-author publications in SCIE journals and international conferences.
- **Production, Evaluation & Security:** LLM-as-a-Judge evaluation (Ragas, DeepEval), input/output guardrails, prompt-injection defense, semantic caching, and token and cost tracking.

Education

Ph.D. in AI Convergence Engineering 2020–2025
Gyeongsang National University, Republic of Korea
Thesis: *Advancements in Soccer Analytics: Players' Goals and Salary Prediction Models with Machine Learning*
Supervisor: Jaeho Kim

Master of Informatics 2018–2020
Gyeongsang National University, Republic of Korea

Bachelor of Mathematics 2014–2018
Fergana State University, Uzbekistan

Experience

AI Engineer & Team Lead Dec 2025–Present
TripleH, Seoul, South Korea

- Lead the AI engineering team designing and shipping production-grade autonomous agent systems.
- Architected and delivered the following AI products:

  - **VIP Agent** — Centralized AI Orchestration Platform that manages and controls a network of specialized sub-agents (market reports, financial RAG brain, KakaoTalk chatbot) from a unified control hub.
  - **Law Agent** — Smart Legal Chatbot implementing Mixture of Decisions: routes queries across multiple LLMs simultaneously, aggregates results, and applies a citability filter for Korean Supreme Court precedents.
  - **Meeting Agent** — Smart Meeting Agent powered by Voice Recognition: real-time on-premise audio transcription, automatic structuring into agenda-based minutes, and 4-format export (PDF, DOCX, Markdown, JSON).
  - **Enterprise System** — On-premise deployment platform enabling all agents to run entirely locally without external API calls, ensuring full data privacy and compliance for enterprise clients.
  - **Digital Workers** — Per-employee AI agents (Digital Twins) that learn each worker's tasks, communication style, and workflows, then act autonomously on their behalf.
- Drive end-to-end development from research and prototyping to on-premise and cloud deployment.
- Establish engineering standards, conduct code reviews, and mentor junior engineers on LLM integration and agent design.

Independent Researcher in Sports Analytics 2025–2026
Self-directed Research Projects

- Conduct independent research on the applications of LLMs and Deep Learning in soccer analytics.
- Develop models for performance evaluation, salary prediction, and tactical analysis using AI.
- Prepare and submit publications to SCIE and AI-focused journals.

Independent AI Educator (Machine Learning, Deep Learning, NLP, LLM) 2023–2026
Online and Private Instruction

- Design and deliver advanced courses in AI, Machine Learning, Deep Learning, and Large Language Models.
- Mentor students and professionals on building end-to-end AI systems and research-oriented projects.
- Focus on practical applications of AI in industry and academia.

Graduate Researcher 2022–2025
Big Data Systems Software Lab, Gyeongsang National University, South Korea

- Focused on ML applications in soccer analytics.
- **Publications:**
  - Malikov, D., et al. *Predicting Soccer Player Salaries with Both Traditional and Automated Machine Learning Approaches* (first author). SCIE: 17% increased accuracy of prediction model. [DOI]
  - Malikov, D., et al. *Beyond xG: A Dual Prediction Model for Analyzing Player Performance Through Expected and Actual Goals in European Soccer Leagues* (first author). SCIE: 12% improvement on player performance prediction. [DOI]
  - Malikov, D., et al. *Comparative Evaluation of Machine Learning Models for Predicting Soccer Injury Types* (first author). KSCIE: Introduced soccer players injury prediction model using constructed features. [DOI]
  - Malikov, D., et al. *The Application of Machine Learning on the Injury Prediction of Soccer Players* (first author). ISE@APSEC: Novel approach for predicting players injury without GPS technology. [PDF]

Graduate Researcher 2018–2021
Distributed Systems Lab, Gyeongsang National University, South Korea

- Focused on applying advanced mathematical concepts to distributed systems.
- **Publications:**
  - On the construction and properties of lattice-group structure in Cartesian product spaces (first author). *Journal of Computer Science*. [DOI]

Mathematics Instructor (University Preparation, SAT & International Math Exams) 2014–2017
Fergana, Uzbekistan

- Taught advanced mathematics and analytical reasoning to high school graduates preparing for university entrance exams, the SAT, and other international mathematics exams.
- Developed customized teaching materials to strengthen students' problem-solving and logical thinking skills.

Projects

LLM Course Assistant — Post-Training Qwen2.5-0.5B (SFT, LoRA, QLoRA) [GitHub]

- Post-trained the pretrained Qwen2.5-0.5B base model into an Uzbek-language chat assistant for students of an LLM course (supervised ML through LLMs, post-training, and RAG).
- **Spec-first workflow:** a written model spec defines language, answer style, tasks, limits, and the success test; training data is written to follow it and every model is judged against it.
- **SFT dataset:** 1,189 training and 63 validation chat examples built from curated seeds, translated instruction data, and generated arithmetic and translation pairs; concepts that appear in the test set are filtered out of training to prevent leakage.
- **LoRA (laptop CPU):** rank 16 on attention and MLP layers, 8.8M of 503M parameters trainable (1.75%), 3 epochs in 151 minutes; validation loss 2.09 → 1.62.
- **QLoRA (Kaggle GPU):** 4-bit NF4 base with double quantization, rank 64; validation loss reached its minimum (1.48) at epoch 3 and rose afterwards, so only the best epoch was kept.
- **Evaluation:** 68 held-out questions across 8 task types with an automatic checker. Base model 5/68 → 32/68 for both adapters; lists 3/10 → 10/10, arithmetic 0/12 → 7/12, saying "I don't know" 0/5 → 4/5.
- **Training details:** loss computed on assistant tokens only; the stop token was chosen after finding that the base model's chat end-token embedding was untrained.
- **Serving:** local FastAPI chat UI with token streaming and live switching between the base model and each LoRA adapter.
- Stack: Python, PyTorch, Hugging Face Transformers, PEFT, bitsandbytes, FastAPI, Kaggle.

Soccer Player Salary Predictor — Industry-level ML Pipeline [GitHub] · **Live**

- Built an end-to-end supervised ML pipeline predicting professional soccer players' annual gross salaries from season-level performance statistics across Bundesliga, La Liga, and Serie A (8 seasons, 2014/15–2021/22).
- **Data leakage prevention:** Train/test split before any fitting; StandardScaler and OrdinalEncoder fitted on training data only; target salary never imputed at any stage.
- **Modeling tiers:** Baseline (Linear Regression, Ridge, Lasso, Decision Tree) → Improved (Random Forest, XGBoost, LightGBM, Gradient Boosting) → Ensemble (Stacking with Ridge meta-learner, Voting regressor).

- **Feature engineering:** 21 derived features without using the target — per-game rates (xG/game, assists/game), efficiency index $(goals+assists)/(xG+xA)$, contribution index, shot conversion, position simplification.
- **Hyperparameter tuning:** Optuna Bayesian optimisation (TPE sampler, 60 trials) over n_estimators, max_depth, learning_rate, subsample, and L1/L2 regularisation.
- **Explainability:** SHAP TreeExplainer — global beeswarm and bar charts, local waterfall and force plots, engineered-feature impact report confirming FE added signal.
- **Deployment:** Gradio demo live on Render, kept active 24/7 via UptimeRobot (5-min ping). Full end-to-end daily logging throughout the pipeline.
- Linked to SCIE publication (Applied Sciences, 2025) — 17% improvement in prediction accuracy. [DOI]
- Stack: Python, XGBoost, LightGBM, Scikit-learn, SHAP, Optuna, Gradio, Render, UptimeRobot.

TacticalDNA — Unsupervised Player-Archetype Discovery [GitHub] · **Live**

- Companion project to the Salary Predictor: same Big-5-league dataset (Bundesliga, La Liga, Serie A), this time solving an unsupervised problem with zero salary/target leakage.
- Built an end-to-end clustering pipeline over 6,314 qualified player-seasons ($\geq$900 minutes) using only per-90 performance features — no salary, age, or team involved in the grouping.
- **Validated finding:** a blind stat line (25 goals, 15 assists, no name attached) was matched by the model to prime Messi's 2019/20 season at 98% cosine similarity — pure identity recovery from raw performance shape, with no player identity ever fed to the model.
- **Algorithm bake-off:** KMeans, Agglomerative (Ward), DBSCAN, and Gaussian Mixture compared on Silhouette, Davies-Bouldin, and Calinski-Harabasz scores; KMeans won outright.
- **Documented negative result:** DBSCAN collapses to a single dense cluster across every eps/min_samples tested — reported as a genuine empirical finding, not hidden.
- **Model selection trade-off:** chose $k = 6$ over the statistically optimal $k = 3$ for product interpretability — a deliberate statistical-optimum-vs-usefulness decision.
- **Archetype naming:** rule-based labeling from cluster centroids — Elite Complete Forward, Clinical Poacher, Deep-Lying Playmaker, Efficient Impact Player, Volume Forward, Defensive Anchor.
- **Deployment:** interactive Streamlit demo on Render; PCA(2) projection chosen over t-SNE/UMAP for stable, real-time inference on new player input.
- Stack: Python, Scikit-learn, Streamlit, Plotly, Render.

Palov — Halal Uzbek Food Platform for Korea **Live**

- Built a full-stack food ordering platform connecting the Uzbek community in South Korea with authentic halal cuisine.
- Implemented multi-country phone authentication (+82 KR / +998 UZ) with OTP verification and Telegram OAuth.
- **AI implementation:** Integrated a Retrieval-Augmented Generation (RAG) system over the menu database so the AI chatbot answers natural-language dish queries with real-time, context-aware recommendations.
- **Self-improvement loop:** User feedback and order patterns are collected and fed back into the recommendation model, allowing the AI to continuously improve suggestion quality over time.
- Stack: Next.js, LLM API, RAG, Vercel.

Hamyon — Personal Finance Manager (*"Pulingiz nazoratda"* — Your money under control) **Live**

- Built real-time expense tracking, budget monitoring, and category-based spending analy-

sis.
- **AI implementation:** Integrated a RAG-powered AI chatbot over the user's financial transaction history, enabling natural-language queries (e.g. "How much did I spend on food this month?") and AI-generated budget insights.
- **Self-improvement loop:** The AI learns from user corrections and preference signals to continuously refine its categorization accuracy and financial advice quality.
- Stack: Next.js, Supabase, LLM API, RAG, Vercel.

Kotib — AI-Powered Personal Secretary (*"Aqlli shaxsiy kotib"*) **Live**

- Built the full AI assistant and chatbot core: natural-language task creation, scheduling, and intelligent note management.
- **AI implementation:** Designed a multi-step agent pipeline — user message → intent classification → action routing (create task / set reminder / generate note / query history) — powered by an LLM with Graph RAG over the user's personal knowledge base.
- **Self-improvement loop:** Tracks completed vs. missed tasks, analyses usage patterns, and autonomously adjusts proactive suggestions and reminder timing based on observed behaviour.
- Implemented persistent conversational memory maintaining full context across all sessions.
- Stack: Next.js, LLM API, Graph RAG, Vercel.

Tools & Frameworks

- **LLM Fine-Tuning & Adaptation:** Supervised Fine-Tuning (SFT), LoRA, QLoRA, Direct Preference Optimization (DPO), instruction tuning, continual pre-training, iterative feedback-driven training pipelines
- **Model Serving & Inference Optimization:** vLLM and TGI inference serving (PagedAttention), dynamic LoRA adapter hot-swapping, quantization (AWQ, GPTQ, GGUF), speculative decoding, local LLM deployment (Ollama, llama.cpp)
- **RAG & Vector Engineering:** Standard RAG, GraphRAG (knowledge graph integration), hybrid search (dense vector + sparse BM25), cross-encoder reranking (Cohere, BGE), hierarchical chunking and parent-child retrieval, vision-based layout parsing (LlamaParse, Unstructured.io), vector databases (pgvector, Qdrant, Milvus)
- **Agentic AI & Orchestration:** LangChain, LangGraph, multi-agent architectures (Supervisor, Router, Network), Mixture of Agents (MoA), plan-and-execute frameworks, self-correction and reflection loops, human-in-the-loop state persistence, tool use (Model Context Protocol, function calling)
- **Production, Evaluation & Security:** LLM-as-a-Judge evaluation (Ragas, DeepEval), LangSmith observability and tracing, input/output guardrails (NeMo Guardrails, Llama Guard), prompt injection and jailbreak defense, semantic caching (GPTCache), token tracking and cost optimization
- **Self-Improving AI Systems:**
  - *Feedback Collection:* Human feedback pipelines (thumbs up/down, ratings, corrections), preference data collection, user behaviour logging
  - *RLHF & Alignment:* Reinforcement Learning from Human Feedback (RLHF), reward model training, Proximal Policy Optimization (PPO), Direct Preference Optimization (DPO)
  - *Iterative Fine-Tuning:* Converting model outputs + human corrections into new training data; continual/online learning loops; automated retraining triggers
  - *Self-Reflection & Critique:* Chain-of-Thought self-correction, Constitutional AI (model critiques and revises its own outputs), LLM-as-Judge automated evaluation
  - *Knowledge & Memory Updates:* RAG knowledge base refresh from new interactions, Graph RAG graph expansion, persistent long-term conversational memory

  - *Prompt Optimization:* Behaviour-based prompt adjustment, automated prompt engineering (DSPy-style), A/B testing of prompt variants
  - *Synthetic Data Generation:* LLM-generated training examples, self-play data creation, data augmentation for low-resource tasks
  - *Active Learning:* Model flags low-confidence outputs for human review; uncertainty sampling to minimise labelling cost
- **Model Hubs & Testing:** Hugging Face (Transformers, PEFT, datasets, model hub), NVIDIA (CUDA, GPU-accelerated inference, NGC model registry, TensorRT)
- **ML / DL Frameworks:** Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy
- **Backend & DevOps:** FastAPI, Docker, GitHub Actions, SQL, Supabase, PostgreSQL (pgvector), Vercel, Render, UptimeRobot
- **Frontend & Web:** Next.js, React, HTML, CSS, JavaScript
- **Computer Vision:** YOLO, Roboflow, OpenCV
- **Research & Writing:** Overleaf, LaTeX

Competences

- **Agentic AI Engineering:** End-to-end design and deployment of multi-agent and Mixture-of-Agents systems for enterprise use cases.
- **LLM & RAG Engineering:** Taking LLM systems from fine-tuning and retrieval design through evaluation, guardrails, and production serving.
- **Team Leadership:** Leading engineering teams, conducting code reviews, and setting technical direction.
- **Research and Continuous Learning:** Eager to explore new AI tools and create models that contribute to science and industry.
- **Collaborative Development:** Works effectively in team environments using Git and GitHub.
- **Presentations:** Skilled at presenting complex ideas clearly to both technical and non-technical audiences.
- **Data Manipulation:** Proficient in SQL for handling and processing large datasets.

Languages

English (Fluent), Uzbek (Native), Turkish (Advanced), Russian (Intermediate), Korean (Conversational)